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METHOD AND DEVICE FOR TRANSMITTING OR RECEIVING DATA DURING SCALING IN WIRELESS COMMUNICATION SYSTEM

Abstract

A method of operating a first distributed unit (DU) in a wireless network system includes monitoring radio resource allocation status information allocated to the first DU, determining changed radio resource allocation information based on the radio resource allocation status information, identifying synchronization time information for applying the changed radio resource allocation information, transmitting, to a second DU, a synchronization message including the synchronization time information and the changed radio resource allocation information, and performing, based on the synchronization message, transmission or reception of a message with a radio unit (RU) that is connected to the first DU and that is connected to the second DU.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of International Application No. PCT/KR2023/017558, filed on Nov. 3, 2023, in the Korean Intellectual Property Receiving Office, which is based on and claims priority to Korean Patent Application No. 10-2023-0073727, filed on Jun. 8, 2023, in the Korean Intellectual Property Office, and Korean Patent Application No. 10-2022-0184835, filed on Dec. 26, 2022, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND

- Field
- [0002] The disclosure relates to a method and device for transmitting and receiving data during scaling in a wireless communication system.
- Description of Related Art
- [0003] In related-art radio access networks (RANs), distributed units (DUs) and cell sites are connected in a 1:N configuration, where the processing capacity of the DU is determined based on the maximum traffic that the cell site may handle. However, the time during which maximum traffic occurs is limited, and outside of this time, a significant number of DU resources may remain unused. Therefore, the resources of DUs may not be efficiently used.
- [0004] Virtualized DU (vDU) pooling may refer to a technology that may reduce the number of servers required to build a RAN system by decoupling the related-art 1:N relationship between DUs and cell sites (a set of radio units (RUs)) and pooling and virtualizing the DUs (referred to as capital expenditure (CAPEX) reduction)). For example, before vDU pooling, four DUs, that is, four servers, are required, but after vDU pooling, it

may be possible to process traffic coming from four cell sites using only three physical servers. As such, with vDU pooling, only three physical servers actually operate, and thus, power consumption may also be saved compared to the related-art RAN system using four physical servers (referred to as operational expenditure (OPEX) reduction)).

[0005] While the above example describes only pooling of DUs, centralized units (CUs) may also be pooled like DUs. A RAN system that lowers CAPEX and OPEX by pooling and virtualizing DUs and CUs in this manner is referred to as a virtualized RAN (vRAN) system.

[0006] There is a growing need for a method for efficiently transmitting and receiving data in vRAN environments.

SUMMARY

[0007] According to an aspect of the disclosure, a method of operating a first distributed unit (DU) in a wireless network system may include monitoring radio resource allocation status information allocated to the first DU, determining changed radio resource allocation information based on the radio resource allocation status information, identifying synchronization time information for applying the changed radio resource allocation information, transmitting, to a second DU, a synchronization message including the synchronization time information and the changed radio resource allocation information, and performing, based on the synchronization message, transmission or reception of a message with a radio unit (RU) that is connected to the first DU and that is connected to the second DU.

[0008] The synchronization message may include common signal indication information, the synchronization time information may include at least one of frame information for applying the changed radio resource allocation information or slot information for applying the changed radio resource allocation information, and the common signal indication information may indicate whether the first DU is to process a common signal.

[0009] The changed radio resource allocation information may include bitmap information indicating at least one second DU resource allocated to the second DU, and indicating at least one first DU resource allocated to the first DU.

[0010] The changed radio resource allocation information may include information about at least one resource region allocated to the first DU or the second DU, and the information about the at least one resource region may include information about a number resource regions, and position information about points constituting the at least one resource region.

[0011] The changed radio resource allocation information may include a reference value corresponding to an integer value that separates resources allocated to the first DU and resources allocated to the second DU.

[0012] The synchronization time information may correspond to preset time information, and the preset time information may include at least one of frame information that is preset for synchronization or slot information that is preset for synchronization.

[0013] The synchronization time information may be identified based on information about a time corresponding to a packet internet groper (PING) message being transmitted from the first DU to the second DU, and information about a time corresponding to a response message to the PING message being transmitted from the second DU to the first DU.

[0014] According to an aspect of the disclosure, a method of operating a second DU in a wireless network system may include receiving, from a first DU, a synchronization message including changed radio resource allocation information and synchronization time information for applying the changed radio resource allocation information, and performing, based on the synchronization message, transmission or reception of a message with a radio unit (RU) that is connected to the first DU and that is connected to the second DU, where the synchronization time information for applying the changed radio resource allocation information may be identified based on radio resource allocation status information allocated to the first DU.

[0015] The synchronization message may include common signal indication information, the synchronization time information may include at least one of frame information for applying the changed radio resource allocation information or slot information for applying the changed radio resource allocation information, and the common signal indication information may include whether the first DU is to process a common signal.

[0016] The changed radio resource allocation information may include bitmap information indicating at least one second DU resource allocated to the second DU, and indicating at least one first DU resource allocated to the first DU.

[0017] The changed radio resource allocation information may include information about at least one resource region allocated to the first DU or the second DU, and the information about the at least one resource region may include information about a number of resource regions, and position information about points constituting the at least one resource region.

[0018] The changed radio resource allocation information may include a reference value corresponding to an integer value that separates resources allocated to the first DU and resources allocated to the second DU.

[0019] The synchronization time information may correspond to preset time information, and the preset time information may include at least one of frame information that is preset for synchronization or slot information that is preset for synchronization.

[0020] The method may include identifying the synchronization time information for applying the changed radio resource allocation information, where the identifying of the synchronization time information may include receiving, from the first DU, a request message including information about a current time resource processed by the first DU, identifying the synchronization time information based on the information about the current time resource processed by the first DU and information about a current time resource processed by the second DU, and transmitting, to the first DU, the synchronization time information based on the request message.

[0021] According to an aspect of the disclosure, a first distributed unit (DU) in a wireless network system, the first DU including: a transceiver; and at least one processor operatively connected to the transceiver, wherein the at least one processor is configured to monitor radio resource allocation status information allocated to the first DU, determine changed radio resource allocation information based on the radio resource allocation status information, identify synchronization time information for applying the changed radio resource allocation information, control the transceiver to transmit, to the second DU, a synchronization message including the synchronization time information and the changed radio resource allocation information, and perform, based on the synchronization message, transmission or reception of a message with the RU.

[0022] The synchronization message may include common signal indication information, the synchronization time information may include at least one of frame information for applying the changed radio resource allocation information and slot information for applying the changed radio resource allocation information, and the common signal indication information may indicate whether the first DU is to process a common signal.

[0023] The changed radio resource allocation information may include bitmap information indicating at least one second DU resource allocated to the second DU, and indicating at least one first DU resource allocated to the first DU.

[0024] The changed radio resource allocation information may include information about at least one resource region allocated to the first DU or the second DU, and the information about the at least one resource region may include information about a number resource regions, and position information about points constituting the at least one resource region.

[0025] The changed radio resource allocation information may include a reference value corresponding to an integer value that separates resources allocated to the first DU and resources allocated to the second DU.

[0026] The synchronization time information may correspond to preset time information, and the preset time information may include at least one of frame information that is preset for synchronization and slot information that is preset for synchronization.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0027] The above and other aspects, features, and advantages of certain embodiments of the present disclosure will be more apparent from the following description taken in conjunction with the accompanying drawings, in which:

[0028] FIG. 1 is a diagram illustrating a wireless communication system according to one or more embodiments;

[0029] FIG. 2 is a diagram illustrating a method of operating a first distributed unit (DU), according to one or more embodiments;
 [0030] FIG. 3 is a flowchart of a method of operating a first DU, according to one or more embodiments;
 [0031] FIG. 4 is a flowchart of a method of operating a second DU, according to one or more embodiments;
 [0032] FIG. 5 is a diagram illustrating message transmission/reception of a first DU, according to one or more embodiments;
 [0033] FIG. 6 is a diagram illustrating a method, performed by a first DU and a second DU, of operating based on changed radio resource allocation information and synchronization time information, according to one or more embodiments;
 [0034] FIG. 7 is a diagram illustrating message transmission/reception of a second DU, according to one or more embodiments;
 [0035] FIG. 8A is a diagram illustrating downlink radio resources according to one or more embodiments;
 [0036] FIG. 8B is a diagram illustrating uplink radio resources according to one or more embodiments;
 [0037] FIG. 9 is a diagram illustrating a structure of bitmap-type changed radio resource allocation information, according to one or more embodiments;
 [0038] FIG. 10 is a diagram illustrating a structure of changed radio resource allocation information including information about allocated resource regions, according to one or more embodiments;
 [0039] FIG. 11 is a diagram illustrating a structure of changed radio resource allocation information including a reference value, according to one or more embodiments;
 [0040] FIG. 12 is a diagram illustrating synchronization time information including preset time information, according to one or more embodiments;
 [0041] FIG. 13 is a diagram illustrating dynamically identified synchronization time information according to one or more embodiments;
 [0042] FIG. 14 is a diagram illustrating synchronization time information determined by a second DU in response to a request from a first DU, according to one or more embodiments.
 [0043] FIG. 15 is a block diagram illustrating a configuration of a first DU; according to one or more embodiments
 [0044] FIG. 16 is a block diagram illustrating a configuration of a second DU, according to one or more embodiments.

DETAILED DESCRIPTION

[0045] Hereinafter, example embodiments of the disclosure will be described in detail with reference to the accompanying drawings. The same reference numerals are used for the same components in the drawings, and redundant descriptions thereof will be omitted. The embodiments described herein are example embodiments, and thus, the disclosure is not limited thereto and may be realized in various other forms. It is to be understood that singular forms include plural referents unless the context clearly dictates otherwise. The terms including technical or scientific terms used in the disclosure may have the same meanings as generally understood by those skilled in the art.

[0046] As used herein, the expression “at least one of a, b, or c” may indicate only a, only b, only c, both a and b, both a and c, both b and c, all of a, b, and c, or variations thereof.

[0047] Although the terms used herein are selected from among common terms that are currently widely used in consideration of their functions the terms may be different according to an intention of one of ordinary skill in the art, a precedent, or the advent of new technology. In addition, in certain cases, there are also terms arbitrarily selected by the applicant, and in this case, the meaning thereof will be defined in detail in the description. Therefore, the terms used herein are not merely designations of the terms, but the terms are defined based on the meaning of the terms and content throughout the present disclosure.

[0048] The singular expression may also include the plural meaning as long as it is not inconsistent with the context. All the terms used herein, including technical and scientific terms, may have the same meanings as those generally understood by those of skill in the art related to the present specification. In addition, although the terms such as ‘first’ or ‘second’ may be used in the present specification so as to describe various elements, these elements should not be limited by the terms. These terms are only used to distinguish one element from another element.

[0049] Throughout the present specification, when a part “includes” a component, it means that the part may additionally include other components rather than excluding other components as long as there is no particular opposing recitation. In addition, as used herein, the terms such as “. . . er (or)”, “. . . unit”, “. . . module”, etc., denote a unit that performs at least one function or operation, which may be implemented as hardware or software or a combination thereof.

[0050] In addition, terms such as “unit” and “module” described in the specification may indicate a unit that processes at least one function or operation, and this may be implemented as hardware or software, or may be implemented as a combination of hardware and software.

[0051] Operations of a method may be performed in an appropriate order unless explicitly described in terms of order. In addition, the use of all illustrative terms (e.g., etc.) is merely for describing technical ideas in detail, and the scope is not limited by these examples or illustrative terms unless limited by the claims.

[0052] FIG. 1 is a diagram illustrating a wireless communication system according to one or more embodiments.

[0053] The wireless communication system may include a core network **110** and a radio access network (RAN) **120**.

[0054] The RAN **120** may include one or more radio units (RUs) **160-1**, **160-N**, **162-1**, **162-N**, **164-1**, and **30**, one or more distributed units (DUs) **150**, **152**, **10**, and **20**, and a centralized unit (CU) **140**. The RAN **120** may include a virtualized RAN (vRAN) system, but is not limited thereto. For example, the RAN **120** may include a 5G system (5GS), a 4G system (4GS), or other wireless communication systems, and may also refer to wireless communication systems developed in the future.

[0055] The CU **140** may refer to an entity that performs functions of some of protocol layers of a network. For example, the CU **140** may refer to an entity that performs network functions of a radio resource control (RRC) layer and a packet data convergence protocol (PDCP) layer. However, the functions of some layers that may be processed by the CU **140** are not limited to the above-described RRC layer and PDCP layer. For example, the CU **140** may refer to a virtualized CU (vCU) in the vRAN system, but is not limited thereto.

[0056] One CU **140** may be connected to N DUs, where N may be any integer greater than 1. The CU **140** and the DUs may be connected to each other via an interface. For example, an interface between the CU **140** and a first DU **10** may be an F1 interface.

[0057] The RAN **120** may include a DU A **150**, a DU B **152**, and the first DU **10**. The DU A **150** and the DU B **152** may perform the same function as the first DU **10**, but the descriptions below will focus on the first DU **10** as an example.

[0058] The first DU **10** may refer to an entity that performs functions of some of protocol layers of the network, excluding some layers performed by the CU **140**. For example, the first DU **10** may refer to an entity that performs network functions (e.g., baseband functions) of a radio link control (RLC) layer, a medium access control (MAC) layer, and a physical (PHY) layer. However, the functions of some layers that may be processed by the first DU **10** are not limited to the above-described RLC layer, MAC layer, and PHY layer. For example, the first DU **10** may refer to a virtualized DU (vDU) in the vRAN system, but is not limited thereto. For example, the first DU **10** may correspond to a processing pod, a component module, any processing operation unit, any deployment unit performing processing, software, etc. For example, the first DU **10** may correspond to one server.

[0059] The DU A **150** may be connected to an RU A-1 **160-1** and an RU A-N **160-N**, and the DU B **152** may be connected to an RU B-1 **162-1** and an RU B-N **162-N**.

[0060] One first DU **10** may be connected to N RUs, where N may be any integer greater than 1. The first DU **10** may be connected to a first RU **164-1** and an RU **30** via interfaces, and for example, the interface between the first DU **10** and the RU **30** may be a fronthaul interface.

[0061] The RU A-1 **160-1**, the RU A-N **160-N**, the RU B-1 **162-1**, the RU B-N **162-N**, and the first RU **164-1** may perform the same function as the RU **30**, and the descriptions below will focus on the RU **30** as an example.

[0062] The RU **30** may refer to an entity that performs some of the functions of the PHY layer, excluding those performed by the first DU **10**. For example, the first DU **10** may perform a function of a high-PHY layer, and the RU **30** may perform a function of a low-PHY layer. For example, the

RU 30 may refer to a virtualized RU (vRU) in the vRAN system, but is not limited thereto.

[0063] A vDU scaling method that dynamically scales in or scales out the first DU 10 based on current traffic conditions to efficiently use server resources may be used. When scaling is performed, the RAN 120 may further include a second DU 20.

[0064] For example, the first DU 10 may refer to the N-th DU and may correspond to a source vDU. The second DU 20 may refer to the N'-th DU and may correspond to a target vDU. The first DU 10 and the second DU 20 may be connected to each other via an inter-DU interface (Xd interface) 130.

[0065] When the amount of data processed by the first DU 10 increases, a new DU may be used for the processing to prevent excessive traffic processing load on the first DU 10. In this case, the RU 30 associated with a particular cell may be migrated to the new DU.

[0066] Scale-out 180 may refer to the addition of the second DU 20 to a DU pool. For example, when the amount of traffic to be processed by the first DU 10 included in the DU pool increases, the second DU 20 may be added. The addition of the second DU 20 to the DU pool may indicate that a processing pod corresponding to the second DU 20 is added to the DU pool. For example, the addition of the second DU 20 to the DU pool may indicate that an additional server capable of processing data begins operation. When the capacity or performance of the first DU 10 (e.g., an existing server) to process data reaches its limit, the second DU 20 (as a non-limiting example, a server with similar specifications) may be added to the communication system by using scale-out.

[0067] Scale-in 190 may refer to the removal of the second DU 20 included in the DU pool. For example, when the amount of traffic to be processed by the DUs included in the DU pool decreases, the second DU 20 may be removed from the DU pool. The removal of the second DU 20 from the DU pool may indicate that the processing pod corresponding to the second DU 20 is removed. For example, the removal of the second DU 20 from the DU pool may indicate that the server processing data ceases the operation. By using scale-in, the number of servers that are no longer required to operate may be reduced, and resources may be saved.

[0068] FIG. 2 is a diagram illustrating a method of operating a first DU, according to one or more embodiments.

[0069] As described above, the first DU 10 and the second DU 20 may be installed as software within a server. When the data processing amount occurring in the RU 30 connected to the first DU 10 increases, the scale-out 180 may be performed such that user equipments (UEs) (e.g., UEs 170-1, 170-N, 172-1, 172-N, 174-1, 174-N) included in a cell where the data processing amount increases may be processed by the second DU 20. When the scale-out 180 is performed, one RU 30 may have been connected to one or more UEs (e.g., hundreds of UEs), and thus, it may be difficult for all UEs included in the RU 30 previously processed by the first DU 10 to be migrated to the second DU 20 at once. Thus, the first DU 10 and the second DU 20 need to transmit data to the RU 30 while changing allocation of radio resources in consideration of the amount of data to be processed.

[0070] The data processing amount may include a throughput, which is the amount of data transmittable per unit time, and may include a value that increases according to the number of UEs managed or processed by the DU described in the present disclosure. The data processing amount may include the amount of data generated in UEs to be processed by the DU, the amount of data to be processed, the amount of communication, the amount of resources, the number of UEs to be processed, and the like.

[0071] Radio resources 210 may refer to resources configured with a frequency domain and a time domain used for transmission and reception of data, and may also be referred to as air resources, wireless radio resources, or the like, but are not limited to these terms. As more radio resources are allocated, more data may be transmitted and received, and the data transmission and reception rates may increase.

[0072] For example, before the scale-out 180 is performed, the first DU 10 may manage the radio resources 210. In addition, during the scale-out 180, the radio resources 210 may be divided into a 1-1st resource region 220 allocated to the first DU 10, and a 2-1st resource region 230 allocated to the second DU 20. The first DU 10 may transmit and receive data to and from the RU 30 by using the 1-1st resource region 220, and the second DU 20 may transmit and receive data to and from the RU 30 by using the 2-1st resource region 230.

[0073] For example, during the scale-out 180, when data generated from UEs that were processed by the first DU 10 is migrated to the second DU 20 and then processed, a change in allocation of resources that were allocated to the first DU 10 and the second DU 20 may be required.

[0074] The resource region allocated to each DU (e.g., the 1-1st resource region 220 or the 2-1st resource region 230) may increase in proportion to the amount of data to be processed by the corresponding DU. During the scale-out 180, as the amount of data processed by the first DU 10 decreases and the amount of data to be processed by the second DU 20 increases, the 1-1st resource region 220 corresponding to the amount of data processed by the first DU 10 may decrease, and the 2-1st resource region 230 corresponding to the amount of data processed by the second DU 20 may increase. On the contrary, during the scale-out 180, as the amount of data processed by the first DU 10 decreases and the amount of data to be processed by the second DU 20 increases, a 1-2nd resource region 225 that is not allocated to the first DU 10 may increase, and a 2-2nd resource region 235 that is not allocated to the second DU 20 may decrease.

[0075] When a margin region is included between the 1-1st resource region 220 and the 2-1st resource region 230 to prevent the 1-1st resource region 220 and the 2-1st resource region 230 from overlapping with each other, there is an issue of wasting expensive resources. Thus, to allocate the radio resources 210 to the first DU 10 and the second DU 20 without waste, a method of setting changed radio resource allocation information is required.

[0076] The changed radio resource allocation information may refer to resource information generated in consideration of allocating resources to at least two DUs (e.g., the first DU 10 and the second DU 20) according to a change in the amount of data processed by each DU during scaling (e.g., scale-out or scale-in) being performed. The changed radio resource allocation information may be generated such that resources allocated to at least two DUs do not overlap each other and a collision does not occur. In addition, the changed radio resource allocation information may be generated such that resources may be set for each DU in proportion to the amount of data processed by the DU. The changed radio resource allocation information may be referred to as radio resource allocation information. The changed radio resource allocation information will be described below with reference to FIGS. 9 to 11.

[0077] When the 1-1st resource region 220 and the 2-1st resource region 230 overlap each other, a UE using a frequency band of the overlapping region may receive erroneous data or may be unable to receive data. To prevent errors in data transmitted to UEs, a time point when the first DU 10 uses the changed resource region 220 allocated to the first DU 10 and a time point when the second DU 20 uses the changed resource region 230 allocated to the second DU 20 need to precisely coincide with each other. Thus, a method of setting synchronization time information for the first DU 10 and the second DU 20 to apply the changed radio resource allocation information at the same time is required.

[0078] The synchronization time information may include information about a time that is set such that times when at least two DUs (e.g., the first DU 10 and the second DU 20) connected to the same RU 30 transmit messages to the RU 30 or receive messages from the RU 30 by using the changed radio resource allocation information coincide with each other. As a non-limiting example, in a case in which resource times in a wireless communication system is defined by frames and slots, a time point when the first DU 10 uses the changed 1-1st resource region 220 (i.e., a frame and a slot) and a time point when the second DU 20 uses the changed 2-1st resource region 230 (i.e., a frame and a slot) need to coincide with each other. Thus, in this case, the synchronization time information may include frame information and slot information.

[0079] In order for the respective DUs to operate at the same time based on the changed radio resource allocation information, the changed radio resource allocation information and the synchronization time information need to be shared with and configured on the first DU 10 and the second DU 20, which are installed on different servers. For example, synchronization of operations between MAC scheduler modules and synchronization of operations between fronthaul handler modules may be required. The operations of the modules will be described in more detail below with reference to FIG. 5.

[0080] The MAC scheduler module may include a module that performs a function of a scheduler among functions of a MAC layer. The MAC scheduler module may allocate resource blocks in a frequency domain and control allocation of uplink resources and downlink resources for symbols

and slots in a time domain, and may include a module that manages radio resources. The first DU **10** may include a first MAC scheduler module **240**, and the second DU **20** may include a second MAC scheduler module **242**. In order for the first DU **10** and the second DU **20** to use the changed radio resource allocation information at the same time, synchronization of operations of the first MAC scheduler module **240** and the second MAC scheduler module **242** may be required.

[0081] A fronthaul handler module of a DU may generate packets of data to be transmitted to the RU **30**, and may include a module that transmits the generated packets to the RU **30** and receives packets generated from the RU **30**. The first DU **10** may include a first fronthaul handler module **250**, the second DU **20** may include a second fronthaul handler module **252**, and the RU **30** may include a third fronthaul handler module **254**. The fronthaul handler modules operate in conjunction with the MAC scheduler modules to receive shared allocation information about UEs such that the DUs and the RU may transmit and receive only data within a resource region to which the UEs are allocated. For example, the first fronthaul handler module **250** may receive allocation information about UEs from the first MAC scheduler module **240**, such that the first DU **10** may transmit, to the RU **30**, only data within a resource region to which the UEs are allocated. In addition, the second fronthaul handler module **252** may receive allocation information about UEs from the second MAC scheduler module **242**, such that the second DU **20** may transmit, to the RU **30**, only data within a resource region to which the UEs are allocated.

[0082] That the operations of the first DU **10** and the second DU **20** are synchronized may indicate that at least two DUs (e.g., the first DU **10** and the second DU **20**) connected to the same RU **30** transmit messages to the RU **30** or receive messages from the RU **30** by using non-overlapping radio resources based on the changed radio resource allocation information, where times when the at least two DUs connected to the same RU **30** transmit or receive the messages coincide with each other based on the synchronization time information. In this case, the fronthaul handler module **254** of the RU **30** may obtain data by using packets received from the first DU **10** and the second DU **20**, without needing to determine which region is valid among the packets transmitted by the first DU **10** and the second DU **20**.

[0083] Hereinafter, a method, performed by the first DU **10**, of determining changed radio resource allocation information and synchronization time information and generating a message for synchronization such that at least two DUs (e.g., the first DU **10** and the second DU **20**) connected to the same RU **30** may operate in synchronization will be described. The message for synchronization may refer to a message transmitted from the first DU **10** to the second DU **20** such that resources allocated to the respective DUs may be applied in the DUs at the same time. The message for synchronization may be a message including at least one of synchronization time information, changed radio resource allocation information, or common signal indication information. The message for synchronization will be described in more detail below with reference to [Table 1].

[0084] FIG. 3 is a flowchart of a method of operating a first DU, according to one or more embodiments.

[0085] In operation **S310**, the first DU **10** may monitor radio resource allocation status information allocated to the first DU **10**.

[0086] The radio resource allocation status information may refer to status information about radio resources allocated to the first DU **10** in relation to the amount of data currently being processed by the first DU **10**. For example, the radio resource allocation status information may include information about whether a change in radio resource allocation information currently allocated to the first DU **10** and the second DU **20** is required, information about whether the amount of resources allocated to the first DU **10** or the second DU **20** needs to be decreased or increased, information about a change in resources to be allocated to the first DU **10** in relation to a change in the amount of data being currently processed or required to be processed (e.g., a resource change rate or amount), and the like. During scale-out, when a change in the amount of data to be processed by each of the first DU **10** and the second DU **20** occurs, the first DU **10** may determine, based on the radio resource allocation status information, that changed radio resource allocation information is required.

[0087] In operation **S320**, the first DU **10** may determine changed radio resource allocation information based on the radio resource allocation status information.

[0088] In one or more embodiments, the changed radio resource allocation information may include bitmap information indicating at least one second DU resource **910** allocated to the second DU **20**, and at least one first DU resource **920** allocated to the first DU **10**.

[0089] In one or more embodiments, the changed radio resource allocation information may include information about one or more resource regions allocated to the first DU **10** or the second DU **20**. The information about the one or more resource regions may include information about the number of one or more resource regions **1031**, **1032**, **1033**, **1034**, **1035**, and **1036**, and position information about points constituting the one or more resource regions.

[0090] In one or more embodiments, the changed radio resource allocation information may include a reference value **1110** corresponding to an integer value that separates resources allocated to the first DU **10** and resources allocated to the second DU **20**.

[0091] In operation **S330**, the first DU **10** may identify synchronization time information for applying the changed radio resource allocation information.

[0092] In one or more embodiments, the synchronization time information may correspond to preset time information. The preset time information may include at least one of preset frame information or slot information for synchronization.

[0093] In one or more embodiments, the synchronization time information may be identified based on information about a time when a packet internet groper (PING) message is transmitted from the first DU **10** to the second DU **20**, and information about a time when a response message to the PING message is transmitted from the second DU **20** to the first DU **10**.

[0094] In one or more embodiments, the first DU **10** may transmit, to the second DU **20**, a request message including information about a current time resource processed by the first DU **10**. The first DU **10** may receive the synchronization time information from the second DU **20** based on the request message. The synchronization time information may be identified based on the information about the current time resource processed by the first DU **10**, and information about a current time resource processed by the second DU **20**.

[0095] In operation **S340**, the first DU **10** may transmit, to the second DU **20**, a message for synchronization including the synchronization time information and the changed radio resource allocation information.

[0096] In one or more embodiments, the message for synchronization may further include common signal indication information. The common signal indication information may be information indicating whether the first DU **10** is to process a common signal.

[0097] In one or more embodiments, the synchronization time information may include at least one of frame information or slot information for applying the changed radio resource allocation information.

[0098] In operation **S350**, the first DU **10** may perform transmission or reception of a message with the RU **30** connected to the first DU **10** and the second DU **20**, based on the message for synchronization.

[0099] FIG. 4 is a flowchart of a method of operating a second DU **20**, according to one or more embodiments.

[0100] In operation **S410**, the second DU **20** may receive, from the first DU **10**, a message for synchronization including changed radio resource allocation information and synchronization time information for applying the changed radio resource allocation information.

[0101] In one or more embodiments, the synchronization time information for applying the changed radio resource allocation information may be identified based on radio resource allocation status information allocated to the first DU. The synchronization time information may include at least one of frame information or slot information for applying the changed radio resource allocation information.

[0102] In one or more embodiments, the message for synchronization may further include common signal indication information. The common signal indication information may be information indicating whether the first DU **10** is to process a common signal.

[0103] In one or more embodiments, the changed radio resource allocation information may include bitmap information indicating at least one second DU resource **910** allocated to the second DU **20**, and at least one first DU resource **920** allocated to the first DU **10**.

[0104] In one or more embodiments, the changed radio resource allocation information may include information about one or more resource regions

allocated to the first DU **10**, the second DU **20**. The information about the one or more resource regions may include information about the number of one or more resource regions **1031**, **1032**, **1033**, **1034**, **1035**, and **1036**, and position information about points constituting the one or more resource regions.

[0105] In one or more embodiments, the changed radio resource allocation information may include a reference value **1110** corresponding to an integer value that separates resources allocated to the first DU **10** and resources allocated to the second DU **20**.

[0106] In one or more embodiments, the synchronization time information may correspond to preset time information. The preset time information may include at least one of preset frame information or slot information for synchronization.

[0107] In one or more embodiments, the synchronization time information may be identified based on information about a time when a PING message is transmitted from the first DU **10** to the second DU **20**, and information about a time when a response message to the PING message is transmitted from the second DU **20** to the first DU **10**.

[0108] In one or more embodiments, the second DU **20** may identify synchronization time information for applying the changed radio resource allocation information. The second DU **20** may receive, from the first DU **10**, a request message including information about a current time resource processed by the first DU **10**. The second DU **20** may identify the synchronization time information based on the information about the current time resource processed by the first DU **10**, and information about a current time resource processed by the second DU **20**. The second DU **20** may transmit the synchronization time information to the first DU **10** based on the request message.

[0109] In operation **S420**, the second DU **20** may perform transmission or reception of a message with the RU **30** connected to the first DU **10** and the second DU **20**, based on the message for synchronization.

[0110] FIG. **5** is a diagram illustrating message transmission/reception of a first DU, according to one or more embodiments.

[0111] The first DU **10** may include a first scale agent module **510**, a first context synchronizer module **520**, a first schedule coordinator module **530**, a first MAC scheduler module **540**, a first fronthaul flow manager module **550**, and a first fronthaul handler module **560**.

[0112] The first scale agent module **510** may refer to a module that coordinates operations required during scale-out and scale-in operations of the first DU **10**. The first scale agent module **510** may refer to a main module for scale operations.

[0113] The first context synchronizer module **520** may refer to a module that checks the amount of data to be processed by the first DU **10** and performs reporting to the first scale agent module **510**. For example, the first context synchronizer module **520** may refer to a module that identifies and manages the amount of data (e.g., the number of UEs) to be processed by the first DU **10**. The first context synchronizer module **520** may refer to a module that identifies which of the first DU **10** or the second DU **20** needs to process data generated from UEs.

[0114] The first schedule coordinator module **530** may refer to a module for performing scheduling. The first schedule coordinator module **530** may include a module that determines and schedules radio resources allocated to the first DU **10**. The first schedule coordinator module **530** may perform a role of changing radio resource allocation information such that the changed radio allocation information may be applied through the first MAC scheduler module **540**. The first schedule coordinator module **530** may include a module that determines radio resource allocation information and synchronization time information and generates a message for synchronization.

[0115] The first MAC scheduler module **540** may include a module for managing radio resources allocated to the first DU **10**. The first MAC scheduler module **540** may perform the same function as the MAC scheduler module of FIG. **2** described above. The first MAC scheduler module **540** may manage a schedule for using a common signal for UEs corresponding to the first DU **10** and radio resources allocated for each UE.

[0116] The first fronthaul flow manager module **550** may refer to a module that determines and controls the extent to which the first DU **10** is responsible for transmitting and receiving packets.

[0117] The first fronthaul handler module **560** may include a module capable of transmitting and receiving data to and from the RU **30** based on information about resources allocated to the first DU **10**.

[0118] The operation of each module will be described by using the case of FIG. **6** below.

[0119] FIG. **6** is a diagram illustrating a method, performed by a first DU and a second DU, of operating based on changed radio resource allocation information and synchronization time information, according to one or more embodiments.

[0120] For convenience of description, the operations of the first DU **10** and the second DU **20** will be described with reference to a case in which a scale-out operation is being performed and the number of UEs connected to a particular RU **30** within the first DU **10** has decreased from 700 to 500. The above-described case is for convenience of description and the present disclosure is not limited thereto. In addition, although operations below is described as operations of the respective modules, the operation of each module may be performed by the first DU **10** or the second DU **20** including the module.

[0121] In operation **S571**, the first context synchronizer module **520** may report, to the first scale agent module **510**, the amount of data currently being processed. For example, the first context synchronizer module **520** may report the amount of data currently being processed (e.g., the number of UEs being managed by the first DU **10**), to the first scale agent module **510** at a preset interval. For example, each time the amount of data processed by each of the first DU **10** and the second DU **20** is changed (e.g., when UEs that were processed by the first DU **10** are migrated to the second DU **20** to be processed), the first context synchronizer module **520** may report, to the first scale agent module **510**, the amount of data currently being processed (e.g., the number of UEs being managed by the first DU **10**).

[0122] For example, referring to FIG. **6**, as the first DU **10** was managing 700 UEs among UEs of the RU **30**, but 200 UEs among the 700 UEs are migrated to the second DU **20**, and the first scale agent module **510** may detect that the first DU **10** is currently managing only 500 UEs among UEs of the RU **30**. The first context synchronizer module **520** may report, to the first scale agent module **510**, the number of currently managed UEs, i.e., **500**, as the amount of data currently being processed.

[0123] In operation **S572**, the first scale agent module **510** may identify, based on the amount of data being processed, whether changed radio resource allocation information is required. For example, the first scale agent module **510** may identify, based on a predetermined reference data processing amount, whether resource allocation information about resources currently allocated to the first DU **10** needs to be changed. The predetermined reference data processing amount is a reference value for determining whether the resource allocation information about resources allocated to the first DU **10** needs to be changed, and as a non-limiting example, may be set as a number of UEs.

[0124] For example, referring to FIG. **6**, because the number of terminals to be processed by the first scale agent module **510** has decreased from 700 to 500, the first scale agent module **510** may identify that the first DU **10** may operate with a reduced amount of resources, decreasing from the amount corresponding to a 1-1st current resource region **610** (e.g., 70% of the entire radio resources) to the amount corresponding to a 1-1st changed resource region **615** (e.g., 50% of the entire radio resources), and may identify that allocation information about allocated resources needs to be changed.

[0125] In operation **S573**, the first scale agent module **510** may transmit the radio resource allocation status information to the first schedule coordinator module **530** to request a change in the radio resource allocation information. Based on determining that a required data processing amount has decreased, the first scale agent module **510** may transmit the radio resource allocation status information to the first schedule coordinator module **530** to notify that a smaller resource region than the current allocation status may be allocated.

[0126] For example, referring to FIG. **6**, the first scale agent module **510** may transmit the radio resource allocation status information to the first schedule coordinator module **530**, along with information that resources may be allocated by changing to the 1-1st changed resource region **615** (e.g., 50% of the entire radio resources). In addition, the second DU **20** was managing 300 UEs, but 200 UEs have been migrated to the second DU **20**. Thus, the second DU **20** may be allocated an amount of resources that has increased from a 2-1st current resource region **610** (e.g., 30% of the entire radio resources) to a 2-2nd changed resource region **625** (e.g., 50% of the entire radio resources).

[0127] In operation S574, the first schedule coordinator module 530 may request scheduling information from the first MAC scheduler module 540. For example, the scheduling information may include information about a frame and a slot that are currently being processed by the first MAC scheduler module 540.

[0128] In operation S575, the first schedule coordinator module 530 may receive the scheduling information from the first MAC scheduler module 540. For example, the first schedule coordinator module 530 may obtain information about a frame and a slot with which data currently being processed by the first MAC scheduler module 540 is transmitted or received.

[0129] For example, the first schedule coordinator module 530 may obtain (83, 5) (i.e., slot 5 of frame 83) as current time information (e.g., (frame, slot)) being processed by the first MAC scheduler module 540.

[0130] In operation S576, the first schedule coordinator module 530 may generate a message for synchronization.

[0131] In one or more embodiments, the first schedule coordinator module 530 may determine changed radio resource allocation information based on the radio resource allocation status information. The changed radio resource allocation information will be described in detail below with reference to the following drawings.

[0132] In one or more embodiments, the first schedule coordinator module 530 may determine synchronization time information for applying the changed radio resource allocation information. For example, the first schedule coordinator module 530 may determine the synchronization time information for applying the changed radio resource allocation information, based on the scheduling information received from the first MAC scheduler module 540. A method of determining the synchronization time information will be described in detail below with reference to the following drawings.

[0133] Referring to FIG. 6, because the first MAC scheduler module 540 is currently processing data of (83, 5) (i.e., slot 5 of frame 83), synchronization time information for the second DU 20 and the first DU 10 to apply the changed radio resource allocation information at the same time may be determined as (86, 7) (i.e., slot 7 of frame 86), considering the time when a message for synchronization is transmitted to the second DU 20, the time when the synchronization time information and the changed radio resource allocation information are transmitted and set in the second DU 20, a data delay due to a network environment, and the like.

[0134] In one or more embodiments, the first schedule coordinator module 530 may generate a message for synchronization including the synchronization time information and the changed radio resource allocation information. The message for synchronization will be described in detail below with reference to [Table 1] below.

[0135] In operation S577, the first schedule coordinator module 530 may transmit the message for synchronization to the second DU 20. For example, the first schedule coordinator module 530 may transmit the message for synchronization to a second scale agent module 710 of the second DU 20. To prevent resources overlapping with resources allocated to the first DU 10 from being used by the second DU 20 and to transmit and receive data by applying the changed radio resource allocation information at the same time with the first DU 10, the second scale agent module 710 may transmit the received message for synchronization to each module in the second DU 20. The operation of each module of the second DU 20 will be described below with reference to FIG. 7.

[0136] In operation S578, the first schedule coordinator module 530 may transmit the message for synchronization to the first fronthaul flow manager module 550.

[0137] In operation S579, the first fronthaul flow manager module 550 may transmit the message for synchronization to the first fronthaul handler module 560. The first fronthaul handler module 560 may transmit a packet for data to the RU 30 or receive a packet from the RU 30 by applying the changed radio resource allocation information at the same time with the second DU 20, without using, by the first DU 10, resources overlapping with resources allocated to the second DU 20.

[0138] For example, the first schedule coordinator module 530 may determine changed radio resource allocation information and synchronization time information for applying the changed radio resource allocation information. The first schedule coordinator module 530 may generate a message for synchronization including the changed radio resource allocation information (e.g., resource allocation information indicating that the first DU 10 may use the 1-1st changed resource region 615 and the second DU 20 may use the 2-2nd changed resource region 625) and the synchronization time information (e.g., (86, 7) (i.e., slot 7 of frame 86)).

[0139] Referring to FIG. 6, the first schedule coordinator module 530 may transmit the message for synchronization to the first fronthaul handler module 560 and a second schedule coordinator module 720 of the second DU 20. For packet transmission and reception, the first fronthaul handler module 560 may set the changed radio resource allocation information and the synchronization time information for applying the changed radio resource allocation information. The second schedule coordinator module 720 may transmit the message for synchronization to the second fronthaul handler module 750, to set the changed radio resource allocation information and the synchronization time information for applying the changed radio resource allocation information, for packet transmission and reception.

[0140] In operation S580, the first MAC scheduler module 540 may transmit, to the first schedule coordinator module 530, time information (e.g., frame information and slot information) about data currently being processed by the first MAC scheduler module 540. The first MAC scheduler module 540 may change and transmit the time information (e.g., frame information and slot information) about the data being processed by the first MAC scheduler module 540, until the synchronization time information coincides with the time information about the data being processed.

[0141] In operation S581, based on the synchronization time information coinciding with time information about data to be currently processed, the first schedule coordinator module 530 may request scheduling from the first MAC scheduler module 540 such that the scheduling is performed based on the changed radio resource allocation information. For example, based on the synchronization time information coinciding with the time information about the data to be currently processed, the first schedule coordinator module 530 may process scheduling based on the changed radio resource allocation information.

[0142] In operation S582, based on the synchronization time information coinciding with the time information about the data to be currently processed, the first fronthaul handler module 560 may transmit a message to the RU 30 or receive a message from the RU 30 based on the changed radio resource allocation information.

[0143] Referring to FIG. 6, the first fronthaul handler module 560 may transmit a message to the RU 30 or receive a message from the RU 30 by using the radio resource allocation information before the change, until the current time coincides with the synchronization time information (e.g., slot 7 of frame 86).

[0144] For example, the first fronthaul handler module 560 may generate a packet 630 by using the radio resource allocation information before the change. The first fronthaul handler module 560 may process, as valid data 632, only a packet corresponding to the radio resource allocation information before the change, and transmit a message to the RU 30 or receive a message from the RU 30.

[0145] For example, when the current time coincides with the synchronization time information (e.g., slot 7 of frame 86), the first fronthaul handler module 560 may process, as valid data 634, only a packet corresponding to the changed radio resource allocation information and transmit the generated packet to the RU 30. For example, a region 636 other than the valid data may include a value of 0.

[0146] The order of the above-described operations may be changed, or some operations may be omitted. For example, the order of operations S577 and S578 may be changed or may be performed at the same time. For example, the order of operations S581 and S582 may be changed or may be performed at the same time.

[0147] The operations of the modules included in the second DU 20 will be described in detail below with reference to the following drawings. Descriptions that are already provided above may be omitted.

[0148] FIG. 7 is a diagram illustrating message transmission/reception of a second DU, according to one or more embodiments.

[0149] The second DU **20** may include the second scale agent module **710**, the second schedule coordinator module **720**, a second MAC scheduler module **730**, a second fronthaul flow manager module **740**, and the second fronthaul handler module **750**.

[0150] The second scale agent module **710** may include a module that coordinates operations required during scale-out and scale-in operations of the second DU **20**. The second scale agent module **710** may include a main module for scale operations.

[0151] The second schedule coordinator module **720** may include a module for performing scheduling. The second schedule coordinator module **720** may include a module that determines radio resources to be allocated to the second DU **20** during scheduling. The second schedule coordinator module **720** may perform a role of changing radio resource allocation information such that the changed radio allocation information may be applied through the second MAC scheduler module **730**.

[0152] The second MAC scheduler module **730** may include a module for managing radio resources allocated to the second DU **20**. The second MAC scheduler module **730** may manage a schedule for using a common signal for UEs corresponding to the second DU **20** and radio resources allocated for each UE.

[0153] The second fronthaul flow manager module **740** may include a module that determines and controls the extent to which the second DU **20** is responsible for transmitting and receiving packets.

[0154] The second fronthaul handler module **750** may include a module capable of transmitting and receiving data to and from an RU based on information about resources allocated to the second DU **20**.

[0155] In operation **S761**, the second scale agent module **710** may receive a message for synchronization from the first schedule coordinator module **530**.

[0156] In operation **S762**, the second scale agent module **710** may transmit the message for synchronization to the second schedule coordinator module **720**, and the second schedule coordinator module **720** may set synchronization time information and changed radio resource allocation information.

[0157] In operation **S763**, the second schedule coordinator module **720** may transmit the message for synchronization to the second fronthaul flow manager module **740**.

[0158] In operation **S764**, the second fronthaul flow manager module **740** may transmit the message for synchronization to the second fronthaul handler module **750**, and the second fronthaul handler module **750** may set synchronization time information and changed radio resource allocation information.

[0159] In operation **S765**, the second MAC scheduler module **730** may transmit, to the second schedule coordinator module **720**, time information (e.g., frame information and slot information) about data currently being processed by the second MAC scheduler module **730**. The second MAC scheduler module **730** may change and transmit the time information (e.g., frame information and slot information) about the data being processed by the second MAC scheduler module **730**, until the synchronization time information coincides with the time information about the data being processed.

[0160] In operation **S766**, based on the synchronization time information coinciding with time information about data to be currently processed, the second schedule coordinator module **720** may request scheduling from the second MAC scheduler module **730** such that the scheduling is performed based on the changed radio resource allocation information. For example, based on the synchronization time information coinciding with the time information about the data to be currently processed, the second schedule coordinator module **720** may process scheduling based on the changed radio resource allocation information.

[0161] In operation **S767**, based on the synchronization time information coinciding with the time information about the data to be currently processed, the second fronthaul handler module **750** may transmit a message to the RU **30** or receive a message from the RU **30** based on the changed radio resource allocation information.

[0162] Referring to FIG. **6**, the second fronthaul handler module **750** may transmit a message to the RU **30** or receive a message from the RU **30** by using the radio resource allocation information before the change, until the current time coincides with the synchronization time information (e.g., slot 7 of frame 86).

[0163] For example, the second fronthaul handler module **750** may generate a packet **640** by using the radio resource allocation information before the change. The second fronthaul handler module **750** may process, as valid data **642**, only a packet corresponding to the radio resource allocation information before the change by using the radio resource allocation information before the change, and may transmit a message to the RU **30** or receive a message from the RU **30**.

[0164] For example, when the current time coincides with the synchronization time information (e.g., slot 7 of frame 86), the second fronthaul handler module **750** may process, as valid data **644**, only a packet corresponding to the changed radio resource allocation information and transmit the generated packet to the RU **30**. For example, a region **646** other than the valid data may include a value of 0.

[0165] The order of the above-described operations may be changed, or some operations may be omitted. For example, the order of operations **S766** and **S767** may be changed or may be performed at the same time.

[0166] FIG. **8A** is a diagram illustrating downlink radio resources according to one or more embodiments.

[0167] In one or more embodiments, downlink data may refer to data transmitted from the first DU **10** and the second DU **20** to the RU **30**. Physical channels for transmitting downlink data may include a physical downlink control channel (PDCCH), a physical downlink shared channel (PDSCH), and a physical broadcast channel (PBCH). For example, the PDCCH may refer to a channel for transmitting control information (e.g., downlink control information (DCI)), but is not limited thereto. The PDSCH may refer to a channel for transmitting user data, but is not limited thereto. The PBCH refers to a channel for transmitting messages to be broadcast, and may refer to, for example, a channel for transmitting cell system information, but is not limited thereto.

[0168] In one or more embodiments, resource regions may include a first downlink resource region **810**, which is a resource region allocated to the first DU **10**, and a second downlink resource region **815**, which is a resource region allocated to the second DU **20**. The first downlink resource region **810** may refer to a resource region allocated for the first DU **10** to transmit data to the RU **30** through the PDSCH, and the second downlink resource region **815** may refer to a resource region allocated for the second DU **20** to transmit data to the RU **30** through the PDSCH.

[0169] Downlink data (or signals) may be classified into common signals and UE-specific signals. The common signals may include control resource set (CORESET) #0 **820**, a synchronization signal block (SSB) **822**, a channel state information reference signal (CSI-RS) **824**, and the like, but are not limited thereto.

[0170] In one or more embodiments, resource regions in which a CORESET (e.g., CORESET #1, CORESET #2, or CORESET #3), which is a UE-specific signal, not a common signal, is transmitted may be allocated separately to each of the first DU **10** or the second DU **20**. For example, data corresponding to a resource region **830** among the data of the CORESET may be transmitted by the first DU **10** using the resource region **830**, and data corresponding to a resource region **835** among the data of the CORESET may be transmitted by the second DU **20** using the resource region **835**.

[0171] FIG. **8B** is a diagram illustrating uplink radio resources according to one or more embodiments.

[0172] In one or more embodiments, uplink data may refer to data transmitted from the first RU **30** to the first DU **10** and the second DU **20**. Physical channels for transmitting uplink data may include a physical random-access channel (PRACH), a physical uplink control channel (PUCCH), and a physical uplink shared channel (PUSCH). For example, the PUCCH may refer to a channel for transmitting control information, but is not limited thereto. The PUSCH may refer to a channel for transmitting user data, but is not limited thereto. The PRACH may refer to a channel for transmitting messages for performing random access, but is not limited thereto.

[0173] In one or more embodiments, resource regions may include a first uplink resource region **850**, which is a resource region allocated to the first DU **10**, and a second uplink resource region **855**, which is a resource region allocated to the second DU **20**. The first uplink resource region **850** may refer to a resource region allocated for the first DU **10** to receive data from the RU **30** through the PUSCH, and the second uplink resource region **855** may refer to a resource region allocated for the second DU **20** to receive data from the RU **30** through the PUSCH.

[0174] In the present disclosure, uplink data (or signals) may be classified into common signals and UE-specific signals. The common signals may include PRACHs **862-1** and **862-2**, sounding reference signals (SRSs) **864-1** and **864-2**, but are not limited thereto.

[0175] In one or more embodiments, resource regions for transmitting PUCCHs, which are UE-specific signals, not common signals, may be allocated to different DUs. For example, PUCCH resource #0 **870** may be allocated to the first DU **10**, PUCCH resource #1 **872** may be allocated to the first DU **10**, PUCCH resource #2 **874** may be allocated to the second DU **20**, and PUCCH resource #3 **876** may be allocated to the second DU **20**.

[0176] Information may include first information (SSBs, CSI-RSs, PRACHs, and common PDCCHs) and second information (UE-specific PDCCHs, PDSCHs, and PUSCHs). Common signals may refer to the first information.

[0177] In one or more embodiments, common signals may be transmitted or received by any one of the first DU **10** or the second DU **20**, and a resource region for transmitting or receiving common signals may be allocated to any one of the first DU **10** or the second DU **20**. For common signals, each of the first DU **10** and the second DU **20** may generate a resource map for transmitting common signals based on preset or provided information. Thus, when the first DU **10** and the second DU **20** are notified of which DU is to transmit or receive common signals, the designated DU, either the first DU **10** or the second DU **20**, may transmit or receive common signals in the resource region for transmitting or receiving common signals, without additional information. Common signal indication information to be described below may refer to information indicating which of the first DU **10** and the second DU **20** is to transmit or receive common signals.

[0178] In one or more embodiments, for signals other than common signals, resources may be allocated to each DU by using changed radio resource allocation information and synchronization time information, and a time at which each DU needs to operate may be set based on the allocated resources.

[0179] Hereinafter, a message for synchronization that may include at least one of synchronization time information, common signal indication information, or changed radio resource allocation information will be described in more detail. In one or more embodiments, the first DU **10** may transmit a message for synchronization (NEWRBMap in [Table 1]) to the second DU **20**.

[0180] In one or more embodiments, the message for synchronization may include synchronization time information. The synchronization time information may be expressed as an integer type and may include at least one of frame information or slot information for applying changed radio resource allocation information. For example, referring to [Table 1], the synchronization time information (e.g., the variable 'time' in [Table 1]) may include two integer values (i.e., frame information and slot information). A method of setting synchronization time information will be described in detail below with reference to FIGS. **11** to **13**.

[0181] In one or more embodiments, the message may include common signal indication information. The common signal indication information may refer to information indicating whether the first DU **10** is to process a common signal. For example, referring to [Table 1], when the value of common_signal indicating common signal indication information is True, it may indicate that the first DU **10** is to process a common signal, and when the value of common_signal is False, it may indicate that the second DU **20** is to process a common signal.

[0182] In one or more embodiments, the message may include changed radio resource allocation information. For example, new_ue_air_resource[2], which represents changed radio resource allocation information, may be expressed by using the data structures of [Table 2] to [Table 4]. struct UERBMap new_ue_air_resource[2] contains two pieces of information, and thus may include uplink resource allocation information and downlink resource allocation information.

TABLE-US-00001 TABLE 1 struct NEWRBMap { int type time[2], // frame, slot bool type common_signal, // true or false struct UERBMap new_ue_air_resource[2] // Uplink, Downlink }

[0183] A method of setting changed radio resource allocation information (e.g., UERBMap in [Table 1]) will be described in more detail below with reference to FIGS. **9** to **11**. For example, methods of setting changed radio resource allocation information include a method using a bitmap, a method using a two-dimensional area range, and a method using a threshold for dividing a resource region into upper and lower parts, which will be described below with reference to the following drawings.

[0184] FIG. **9** is a diagram illustrating a structure of bitmap-type changed radio resource allocation information, according to one or more embodiments.

[0185] In one or more embodiments, changed radio resource allocation information may include at least one second DU resource **910** allocated to the second DU **20**, and at least one first DU resource **920** allocated to the first DU **10**.

[0186] In one or more embodiments, the first DU **10** may indicate, for each resource, whether the resource is allocated to the first DU **10** or allocated to the second DU **20**. For example, a resource may correspond to a resource element (RE) or a resource block (RB).

[0187] A resource region may be composed of 14 symbols and 273 REs, but the number of symbols and the number of REs are not limited thereto. The first DU **10** may generate changed radio resource allocation information in the form of a 14×273 bitmap. For example, for a resource region in which 14 symbols and 273 REs exist, the changed radio resource allocation information may include information indicating whether each RE is allocated to the first DU **10** or the second DU **20**. For example, when a bit of the bitmap is 0, it may indicate a resource allocated to the first DU **10**, and when a bit of the bitmap is 1, it may indicate a resource allocated to the second DU **20**. In addition, as another example, when a bit of the bitmap is 0, it may indicate a resource allocated to the second DU **20**, and when a bit of the bitmap is 1, it may indicate a resource allocated to the first DU **10**.

[0188] The 14 symbols represent one slot format, and a pattern may be repeated every 14 symbols. In a resource grid, one symbol may have up to 273 REs. In addition, one symbol may have up to 273 RBs depending on the bandwidth and a subcarrier spacing (SCS) value. Thus, in bitmap-type changed radio resource allocation information (e.g., UERBMap in [Table 2]), each bit may include information indicating whether the corresponding resource, either an RB or an RE, is allocated to the first DU **10** or the second DU **20**.

TABLE-US-00002 TABLE 2 Struct UERBMap { int_16 map[273] }

[0189] Bitmap-type radio resource allocation information may provide an effect of flexible allocation of resources to the first DU **10** and the second DU **20**. Furthermore, the bitmap-type radio resource allocation information allows for allocation of small units of resources to the first DU **10** and the second DU **20**, enabling efficient use of radio resources.

[0190] FIG. **10** is a diagram illustrating a structure of changed radio resource allocation information including information about allocated resource regions, according to one or more embodiments.

[0191] In one or more embodiments, changed radio resource allocation information may include information about one or more resource regions. Referring to FIG. **10**, radio resources may include a second DU region **1010** allocated to the second DU **20**, and a first DU region **1020** allocated to the first DU **10**. For example, the second DU region **1010** and the first DU region **1020** may be configured in units of physical resource blocks (PRBs), RBs, or REs.

[0192] In one or more embodiments, the first DU region **1020** may include a 1-1st DU region **1031**, a 1-2nd DU region **1032**, a 1-3rd DU region **1033**, a 1-4th DU region **1034**, a 1-5th DU region **1035**, and a 1-6th DU region **1036**.

[0193] In one or more embodiments, the 1-1st DU region **1031** may correspond to a quadrangular region formed by a first point **1041**, a second point **1042**, a third point **1043**, and a fourth point **1044**. However, definition of the 1-1st DU region **1031** does not require four points, and the above-

described four values may be varied.

[0194] In one or more embodiments, the first DU **10** may generate radio resource allocation information including the information about the resource region described above by using a PRB MAP pointer.

[0195] [Table 3] below is for describing a structure of radio resource allocation information in which a PRB MAP pointer is used.

[0196] Referring to [Table 3], NumberofArea may indicate the number of at least one or more resource regions to be identified. That is, in the example of FIG. **10**, the value of NumberofArea may be 6.

[0197] In PRB[x], the value of x may represent the number of points to be used to define a resource region. That is, in the example of FIG. **10**, PRB[4] may indicate that resource information about the first point **1041**, resource information about the second point **1042**, resource information about the third point **1043**, and resource information about the fourth point **1044** are to be included. For example, the first point **1041** may represent (first PRB, first symbol), the second point **1042** may represent (first PRB, second symbol), the third point **1043** may represent (second PRB, second symbol), and the fourth point **1044** may represent (second PRB, first symbol). For example, the first symbol may refer to the first symbol constituting the resource region, and the second symbol may refer to the last symbol.

TABLE-US-00003 TABLE 3 Struct UERBMap { int PRB[4][NumberofArea] }

[0198] For example, FIG. **10** illustrates an example in which information about the resource region allocated to the first DU **10** is indicated, but it is also possible to indicate information about the resource region allocated to the second DU **20**. The changed radio resource allocation information including the information about the allocated resource regions has an effect that the resource regions may be allocated to the first DU **10** and the second DU **20** for each two-dimensional region. The changed radio resource allocation information including the information about the allocated resource regions has an effect that the resources may be allocated with a smaller size of data compared to the bitmap format of FIG. **10**. In addition, there is an effect that less complicated resource allocation is possible compared to the bitmap format of FIG. **10**.

[0199] FIG. **11** is a diagram illustrating a structure of changed radio resource allocation information including a reference value, according to one or more embodiments.

[0200] In one or more embodiments, the first DU **10** may generate radio resource allocation information by using the reference value **1110**. The first DU **10** may include, in a message, the reference value **1110** corresponding to a particular value.

[0201] [Table 4] below describes a structure of radio resource allocation information in which the reference value **1110** is used, and 'reference', corresponding to the reference value, has an integer type.

[0202] In one or more embodiments, the reference value **1110** in the message may correspond to an integer corresponding to any one of the numbers from 0 to 273. Here, 0 to 273 may represent index values of REs.

TABLE-US-00004 TABLE 4 Struct UERBMap { int reference }

[0203] Referring to FIG. **11**, a resource region located below the reference value **1110** may be set as a first DU region **1130**. A resource region located above the reference value **1110** may be set as a second DU region **1120**. In one or more embodiments, based on the reference value **1110**, the first DU **10** may use at least one RB having an index value corresponding to a value less than the reference value **1110**. That is, the first DU region **1130** allocated to the first DU **10** may be a resource region located below the reference value **1110**. In one or more embodiments, based on the reference value **1110**, the second DU **20** may use at least one RB having an index value corresponding to a value greater than or equal to the reference value **1110**. That is, the second DU region **1120** allocated to the second DU **20** may be a resource region located above the reference value **1110**.

[0204] In one or more embodiments, a resource region located above the reference value **1110** may be set as the first DU region **1130**, and a resource region located below the reference value **1110** may be set as the second DU region **1120**.

[0205] The structure of the radio resource allocation information using the reference value **1110** has an advantage that the size of data to be shared between the first DU **10** and the second DU **20** is simple.

[0206] In addition, the structure of the radio resource allocation information using the reference value **1110** has an advantage that a cost for generating resource mapping information is low.

[0207] The structure of the radio resource allocation information using the reference value **1110** has an advantage that it may be used when indicating information about radio resources simply allocated between the first DU **10** and the second DU **20**.

[0208] Hereinafter, a method, performed by the first DU **10**, of determining synchronization time information for applying changed radio resource allocation information will be described.

[0209] FIG. **12** is a diagram illustrating synchronization time information including preset time information, according to one or more embodiments.

[0210] The first DU **10** may determine synchronization time information, considering a time taken for a first schedule coordinator **1220** to transmit a message to a second schedule coordinator **1250** of the second DU **20**. Within the first DU **10**, a time required for changed radio resource allocation information to be transmitted to a first fronthaul handler **1210** may be less than a time required for the first DU **10** to transmit the changed radio resource allocation information to the second schedule coordinator **1250**. This is because the transmission of the changed radio resource allocation information to the first fronthaul handler **1210** is an operation that occurs in the same environment in the first DU **10**, but the transmission of the changed radio resource allocation information to the second schedule coordinator **1250** is an operation that occurs in different environments.

[0211] In one or more embodiments, in the first DU **10**, the synchronization time information may include preset time information. The preset time information may be a fixed time value. During system configuration, a communication time required for transmitting and receiving messages between the first DU **10** and the second DU **20** may be measured, and the measured communication time may be used as the preset time information. For example, the preset time information may include at least one of frame information or slot information.

[0212] In one or more embodiments, when the first DU **10** (e.g., a first server) and the second DU **20** (e.g., a second server) are installed, the preset time information may be a value determined based on a time when a message is transmitted from the first schedule coordinator **1220** to a first scale agent **1230**, a time when a message is transmitted from the first scale agent **1230** to a second scale agent **1260**, and a time when a message is transmitted from the second scale agent **1260** to the second schedule coordinator **1250**.

[0213] In one or more embodiments, the first DU **10** may determine synchronization time information for applying the changed radio resource allocation information, by adding the preset time information to a time point when a radio resource change-requiring event occurs by Equation (1) below.

[00001] Synchronizationtimeinformation = Timepointwhenradioresourcechange - requiringeventoccurs + Presetimeinformation (1)

[0214] In one or more embodiments, the synchronization time information may refer to a time for applying the changed radio resource allocation information, and may represent time resource information. As a non-limiting example, the synchronization time information may include at least one of frame information or slot information for applying the changed radio resource allocation information.

[0215] In one or more embodiments, the time point when the radio resource change-requiring event occurs may be a time point when the first DU **10** determines that a change in the changed radio resource allocation information is required. For example, the time point when the radio resource change-requiring event occurs may be a time point when the first DU **10** starts or completes generation of the changed radio resource allocation information. As a non-limiting example, the time point when the radio resource change-requiring event occurs may include at least one of frame information or slot information about the time point when the radio resource change-requiring event occurs.

[0216] FIG. **13** is a diagram illustrating dynamically identified synchronization time information according to one or more embodiments.

[0217] A PING message may refer to a message to transmit an Internet Control Message Protocol (ICMP) echo request to another device, which returns a response message when the PING message is received. A fast response time may indicate a low latency in the network environment through which the PING message has been transmitted.

[0218] In one or more embodiments, the synchronization time information may be identified based on information about a time when a PING message is transmitted from the first DU **10** to the second DU **20**, and information about a time when a response message to the PING message is transmitted from the second DU **20** to the first DU **10**.

[0219] In one or more embodiments, the information about the time when the PING message is transmitted from the first DU **10** to the second DU **20** may refer to a value obtained by summing a 1-1st transmission time **1370-1** when the PING message is transmitted from a first schedule coordinator **1320** to a first scale agent **1330**, a 1-2nd transmission time **1370-2** when the PING message is transmitted from the first scale agent **1330** to a second scale agent **1360**, and a 1-3rd transmission time **1370-3** when the PING message is transmitted from the second scale agent **1360** to a second schedule coordinator **1350**.

[0220] In one or more embodiments, the information about the time when the response message to the PING message is transmitted from the second DU **20** to the first DU **10** may refer to a value obtained by summing a 2-1st transmission time **1380-1** when the response message is transmitted from the second schedule coordinator **1350** to the second scale agent **1360**, a 2-2nd transmission time **1380-2** when the response message is transmitted from the second scale agent **1360** to the first scale agent **1330**, and a 2-3rd transmission time **1380-3** when the response message is transmitted from the first scale agent **1330** to the first schedule coordinator **1320**.

[0221] In one or more embodiments, the synchronization time information may be determined by using Equation (2) below.

[00002]

$$\text{Synchronizationtimeinformation} = \text{Timepointwhenradioresourcechange} - \text{requiringeventoccurs} + \text{InformationabouttimewhenPINGmessageistransmittedfrom}$$

[0222] FIG. **14** is a diagram illustrating synchronization time information determined by a second DU in response to a request from a first DU, according to one or more embodiments.

[0223] A first schedule coordinator **1420** may transmit, to a second schedule coordinator **1450**, a request message **1470** for requesting synchronization time information for applying changed radio resource allocation information, including information about a time resource currently being processed by the first DU **10**. For example, the information about the time resource currently being processed by the first DU **10** may include at least one of frame information or slot information corresponding to the time resource currently being processed. For example, to set a time point for applying the changed radio resource allocation information, the first schedule coordinator **1420** may transmit the request message **1470** including the information about the time resource currently being processed (e.g., information indicating (Frame, Slot)=(273,14)).

[0224] The second schedule coordinator **1450** may determine synchronization time information for applying the changed radio resource allocation information, based on the information about the time resource currently being processed by the first DU **10**, and information about a time resource currently being processed by the second DU **20**. For example, the information about the time resource currently being processed by the second DU **20** may include at least one of frame information or slot information corresponding to the information about the time resource currently being processed by the second DU **20**. For example, the second schedule coordinator **1450** may identify time difference information (e.g., 7 slots) between information about a time resource (e.g., (Frame, Slot)=(273,14)) received from the first DU **10** and the information about the time resource currently being processed by the second DU **20** (e.g., (Frame, Slot)=(274,1)). Based on the time difference information (e.g., 7 slots), the second schedule coordinator **1450** may determine the synchronization time information to be 14 slots later (e.g., (Frame, Slot)=(274,15)), considering the round-trip time of the message.

[0225] The second schedule coordinator **1450** may transmit, to the first schedule coordinator **1420**, synchronization time information **1480** determined by the second DU **20**. For example, the synchronization time information may include at least one of frame information or slot information corresponding to a time resource for applying the changed radio resource allocation information.

[0226] FIG. **15** is a block diagram illustrating a configuration of a first DU according to one or more embodiments.

[0227] The first DU **10** may include a transceiver **1510**, a processor **1520**, and a memory **1530**. The components of the first DU **10** are not limited to the above examples. For example, the first DU **10** may include more or fewer components than the above-described components. In one or more embodiments, the transceiver **1510**, the processor **1520**, and the memory **1530** may be implemented as a single chip. In addition, the processor **1520** may include one or more processors.

[0228] The transceiver **1510** refers to both a receiver and a transmitter of the first DU **10**, and may transmit and receive signals to and from the second DU **20** or the RU **30**. Signals transmitted to and received from the second DU **20** or the RU **30** may include control information and data. To this end, the transceiver **1510** may include a radio-frequency (RF) transmitter that up-converts and amplifies the frequency of a signal being transmitted, and an RF receiver that performs low-noise amplification and down-converts the frequency of a received signal. However, this is merely an example of the transceiver **1510**, and the components of the transceiver **1510** are not limited to the RF transmitter and the RF receiver.

[0229] In addition, the transceiver **1510** may perform functions for transmitting and receiving signals through a radio channel. For example, the transceiver **1510** may receive a signal through a radio channel and output the received signal to the processor **1520**, and may transmit a signal output from the processor **1520** through a radio channel.

[0230] The memory **1530** may store programs and data required for the operation of the first DU **10**. In addition, the memory **1530** may store control information or data included in a signal obtained by the device. The memory **1530** may include a storage medium such as read-only memory (ROM), random-access memory (RAM), a hard disk, a compact disc (CD) ROM (CD-ROM), or a digital video disc (DVD), or a combination of storage media. In addition, the memory **1530** may not be a separate component but may be included in the processor **1520**. The memory **1530** may include a volatile memory, a nonvolatile memory, or a combination of a volatile memory and a nonvolatile memory. In addition, the memory **1530** may provide data stored therein upon a request from the processor **1520**.

[0231] The processor **1520** may control a series of operations to allow the first DU **10** to operate according to the above-described embodiments of the present disclosure. For example, the processor **1520** may receive a control signal and a data signal through the transceiver **1510** and process the received control signal and data signal. The processor **1520** may transmit the processed control signal and data signal through the transceiver **1510**. In addition, the processor **1520** may write or read data to or from the memory **1530**. The processor **1520** may perform functions of a protocol stack required by a communication standard. To this end, the processor **1520** may include at least one processor or microprocessor. In one or more embodiments, the processor **1520** or a part of the transceiver **1510** may be referred to as a communication processor (CP).

[0232] The processor **1520** may include one or more processors. In this case, the one or more processors may be a general-purpose processor, such as a central processing unit (CPU), an application processor (AP), or a digital signal processor (DSP), a dedicated graphics processor such as a graphics processing unit (GPU) or a vision processing unit (VPU), or a dedicated artificial intelligence processor such as a neural processing unit (NPU). For example, in a case in which the one or more processors are dedicated artificial intelligence processors, the dedicated artificial intelligence processor may be designed with a hardware structure specialized for processing a particular artificial intelligence model.

[0233] The at least one processor **1520** may monitor radio resource allocation status information allocated to the first DU **10**.

[0234] In one or more embodiments, the at least one processor **1520** may determine changed radio resource allocation information based on the radio resource allocation status information.

[0235] When time points when the first DU **10** and the second DU **20** apply the changed radio resource allocation information are different from each other, a situation may occur in which both the first DU **10** and the second DU **20** do not use radio resources or use a partial radio resource region at the same time. This leads to communication quality degradation and results in inefficient use of radio resources. According to one or more embodiments, it is possible to provide an effect that radio resources are allocated without wasted resources, considering the amounts of data processed by the first DU **10** and the second DU **20**.

[0236] In one or more embodiments, at least one processor **1520** may identify synchronization time information for applying the changed radio

resource allocation information.

[0237] In one or more embodiments, the synchronization time information corresponds to preset time information, and the preset time information may include at least one of frame information or slot information that is preset for synchronization. According to one or more embodiments, synchronization time information that reflects a delay value in consideration of a network environment in which DUs are installed may be provided.

[0238] In one or more embodiments, the synchronization time information may be identified based on information about a time when a PING message is transmitted from the first DU **10** to the second DU **20**, and information about a time when a response message to the PING message is transmitted from the second DU **20** to the first DU **10**. According to one or more embodiments, synchronization time information that reflects a dynamically measured minimum delay value may be provided.

[0239] In one or more embodiments, the at least one processor **1520** may transmit, to the second DU **20**, a request message including information about a current time resource processed by the first DU **10**. The at least one processor **1520** may receive the synchronization time information from the second DU **20** based on the request message. The synchronization time information may be identified based on the information about the current time resource processed by the first DU **10**, and information about a current time resource processed by the second DU **20**. According to one or more embodiments, synchronization time information considering operations of the second DU **20** and the first DU **10** may be provided.

[0240] In one or more embodiments, the at least one processor **1520** may control the transceiver **1510** to transmit, to the second DU **20**, a message for synchronization including the synchronization time information and the changed radio resource allocation information.

[0241] According to one or more embodiments, for signals other than common signals, the changed radio resource allocation information may be flexibly processed in various forms, and thus may be applied to various systems.

[0242] In one or more embodiments, the changed radio resource allocation information may include bitmap information indicating at least one second DU resource **910** allocated to the second DU **20**, and at least one first DU resource **920** allocated to the first DU **10**. According to one or more embodiments, the bitmap information may provide an effect of enabling diverse and detailed allocation of resources to DUs.

[0243] In one or more embodiments, the changed radio resource allocation information may include information about one or more resource regions allocated to the first DU **10** or the second DU **20**. The information about the one or more resource regions may include information about the number of one or more resource regions **1031**, **1032**, **1033**, **1034**, **1035**, and **1036**, and position information about points constituting the one or more resource regions. According to one or more embodiments, a method using information about one or more resource regions may provide an effect of allocating resources to DUs in various ways, while simultaneously reducing the size of a message for synchronization and the time required for synchronization.

[0244] In one or more embodiments, the changed radio resource allocation information may include the reference value **1110** corresponding to an integer value that separates resources allocated to the first DU **10** and resources allocated to the second DU **20**. According to one or more embodiments, a method using a reference value may provide an effect of significantly reducing the size of a message for synchronization and significantly reducing the time required for synchronization.

[0245] In one or more embodiments, the at least one processor **1520** may perform transmission or reception of a message with the RU **30** connected to the first DU **10** and the second DU **20**, based on the message for synchronization.

[0246] According to one or more embodiments, because the synchronization time information is included for each changed radio resource allocation information, the module performing the corresponding operation may easily and quickly identify when to apply the changed radio resource allocation information, simply by checking the synchronization time information (e.g., the 'time[2]' field) included in the message for synchronization.

[0247] In one or more embodiments, the message for synchronization may further include common signal indication information. The common signal indication information may be information indicating whether the first DU **10** is to process a common signal. According to one or more embodiments, because the message for synchronization includes the common signal indication information, it is not necessary to share information about a common signal, which has a complex structure, between servers (or nodes). That is, encoding and decoding processes for constructing the data structure for information about a common signal may be omitted. In addition, information such as a slot-by-slot schedule pattern for information about a common signal does not need to be shared between DUs.

[0248] In one or more embodiments, the synchronization time information may include at least one of frame information or slot information for applying the changed radio resource allocation information.

[0249] The fronthaul handler module **254** of the RU **30** may not be able to determine which region is valid among packets transmitted by the first DU **10** and the second DU **20**. Thus, the fronthaul handler needs to obtain the current status of air resources used by the MAC scheduler of each DU, in order to transmit or receive necessary packets to or from the RU.

[0250] According to one or more embodiments, it is possible to provide an effect that, during scaling (scale-in or scale-out), the MAC scheduler modules and the fronthaul handler modules of the first DU **10** and the second DU **20** apply the changed radio resource allocation information at the same time.

[0251] According to one or more embodiments, the MAC scheduler modules may prevent the first DU **10** and the second DU **20** from using the same radio resource region, thereby providing stable communication.

[0252] According to one or more embodiments, because there are no unused regions by both the first DU **10** and the second DU **20**, scheduling may be performed without wasting radio resources.

[0253] According to one or more embodiments, the fronthaul handler module may distinguish between data to be transmitted and data to be received according to radio resources, and may apply radio resource information at the same time with the MAC scheduler module, and thus, it is possible to provide an effect that the amount of data to be transmitted and received between the DUs and the RU **30** may be optimized.

[0254] According to one or more embodiments, because the first DU **10** and the second DU **20** transmit, to the RU **30**, only packets corresponding to the positions of radio resources used by the first DU **10** and the second DU **20** based on the changed radio resource allocation information, even when the packets are received as is and then transmitted to UEs, there is no mutual interference between the UEs of the first DU **10** and the second DU **20**.

[0255] FIG. **16** is a block diagram illustrating a configuration of a second DU according to one or more embodiments.

[0256] The second DU **20** may include a transceiver **1610**, a processor **1620**, and a memory **1630**. The components of the second DU **20** are not limited to the above examples. For example, the second DU **20** may include more or fewer components than the above-described components. In one or more embodiments, the transceiver **1610**, the processor **1620**, and the memory **1630** may be implemented as a single chip. In addition, the processor **1620** may include one or more processors.

[0257] The transceiver **1610** refers to both a receiver and a transmitter of the second DU **20**, and may transmit and receive signals to and from the first DU **10** or the RU **30**. Signals transmitted to and received from the first DU **10** or the RU **30** may include control information and data. To this end, the transceiver **1610** may include an RF transmitter that up-converts and amplifies the frequency of a signal being transmitted, and an RF receiver that performs low-noise amplification and down-converts the frequency of a received signal. However, this is merely an embodiment of the transceiver **1610**, and the components of the transceiver **1610** are not limited to the RF transmitter and the RF receiver.

[0258] In addition, the transceiver **1610** may perform functions for transmitting and receiving signals through a radio channel. For example, the transceiver **1610** may receive a signal through a radio channel and output the received signal to the processor **1620**, and may transmit a signal output from the processor **1620** through a radio channel.

[0259] The memory **1630** may store programs and data required for the operation of the second DU **20**. In addition, the memory **1630** may store control information or data included in a signal obtained by the device. The memory **1630** may include a storage medium such as ROM, RAM, a

hard disk, a CD-ROM, or a DVD, or a combination of storage media. In addition, the memory **1630** may not be a separate component but may be included in the processor **1620**. The memory **1630** may include a volatile memory, a nonvolatile memory, or a combination of a volatile memory and a nonvolatile memory. In addition, the memory **1630** may provide data stored therein upon a request from the processor **1620**.

[0260] The processor **1620** may control a series of operations to allow the second DU **20** to operate according to the above-described embodiments of the present disclosure. For example, the processor **1620** may receive a control signal and a data signal through the transceiver **1610** and process the received control signal and data signal. The processor **1620** may transmit the processed control signal and data signal through the transceiver **1610**. In addition, the processor **1620** may write or read data to or from the memory **1630**. The processor **1620** may perform functions of a protocol stack required by a communication standard. To this end, the processor **1620** may include at least one processor or microprocessor. In one or more embodiments, the processor **1620** or a part of the transceiver **1610** may be referred to as a CP.

[0261] The processor **1620** may include one or more processors. In this case, the one or more processors may be a general-purpose processor, such as a CPU, an AP, or a DSP, a dedicated graphics processor such as a GPU or a VPU, or a dedicated artificial intelligence processor such as an NPU. For example, in a case in which the one or more processors are dedicated artificial intelligence processors, the dedicated artificial intelligence processor may be designed with a hardware structure specialized for processing a particular artificial intelligence model.

[0262] In one or more embodiments, the at least one processor **1620** may receive, from the first DU **10**, a message for synchronization including changed radio resource allocation information and synchronization time information for applying the changed radio resource allocation information.

[0263] In one or more embodiments, the synchronization time information for applying the changed radio resource allocation information may be identified based on radio resource allocation status information allocated to the first DU. The synchronization time information may include at least one of frame information or slot information for applying the changed radio resource allocation information.

[0264] In one or more embodiments, the message for synchronization may further include common signal indication information. The common signal indication information may be information indicating whether the first DU **10** is to process a common signal.

[0265] In one or more embodiments, the changed radio resource allocation information may include bitmap information indicating at least one second DU resource **910** allocated to the second DU **20**, and at least one first DU resource **920** allocated to the first DU **10**.

[0266] In one or more embodiments, the changed radio resource allocation information may include information about one or more resource regions allocated to the first DU **10** or the second DU **20**. The information about the one or more resource regions may include information about the number of one or more resource regions **1031**, **1032**, **1033**, **1034**, **1035**, and **1036**, and position information about points constituting the one or more resource regions.

[0267] In one or more embodiments, the changed radio resource allocation information may include a reference value **1110** corresponding to an integer value that separates resources allocated to the first DU **10** and resources allocated to the second DU **20**.

[0268] In one or more embodiments, the synchronization time information may correspond to preset time information. The preset time information may include at least one of preset frame information or slot information for synchronization.

[0269] In one or more embodiments, the synchronization time information may be identified based on information about a time point when a PING message is transmitted from the first DU **10** to the second DU **20**, and information about a time point when a response message to the PING message is transmitted from the second DU **20** to the first DU **10**.

[0270] In one or more embodiments, at least one processor **1620** may identify synchronization time information for applying the changed radio resource allocation information. The at least one processor **1620** may receive, from the first DU **10**, a request message including information about a current time resource processed by the first DU **10**. The at least one processor **1620** may identify the synchronization time information based on the information about the current time resource processed by the first DU **10**, and information about a current time resource processed by the second DU **20**. The at least one processor **1620** may transmit the synchronization time information to the first DU **10** based on the request message.

[0271] In one or more embodiments, the at least one processor **1620** may perform transmission or reception of a message with the RU **30** connected to the first DU **10** and the second DU **20**, based on the message for synchronization.

[0272] In one or more embodiments, a computer-readable recording medium having recorded thereon a program for causing a computer to execute the method of operating the first DU **10** and the method of operating the second DU **20** described above may be provided.

[0273] According to one or more embodiments, methods according to various embodiments disclosed herein may be included in a computer program product and then provided. The computer program product may be traded as commodities between sellers and buyers. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., a CD-ROM), or may be distributed online (e.g., downloaded or uploaded) through an application store or directly between two user devices (e.g., smart phones). In a case of online distribution, at least a portion of the computer program product (e.g., a downloadable app) may be temporarily stored in a machine-readable storage medium such as a manufacturer's server, an application store's server, or a memory of a relay server.

[0274] A machine-readable storage medium may be provided in the form of a non-transitory storage medium. Here, the term 'non-transitory storage medium' refers to a tangible device and does not include a signal (e.g., an electromagnetic wave), and the term 'non-transitory storage medium' does not distinguish between a case where data is stored in a storage medium semi-permanently and a case where data is stored temporarily. For example, the 'non-transitory storage medium' may include a buffer in which data is temporarily stored.

[0275] As used in connection with various embodiments of the disclosure, the term "module" may include a unit implemented in hardware, software, or firmware, and may interchangeably be used with other terms, for example, logic, logic block, part, or circuitry. A module may be a single integral component, or a minimum unit or part thereof, adapted to perform one or more functions. For example, according to an embodiment, the module may be implemented in a form of an application-specific integrated circuit (ASIC).

[0276] Various embodiments as set forth herein may be implemented as software including one or more instructions that are stored in a storage medium that is readable by a machine. For example, a processor of the machine may invoke at least one of the one or more instructions stored in the storage medium, and execute it, with or without using one or more other components under the control of the processor. This allows the machine to be operated to perform at least one function according to the at least one instruction invoked. The one or more instructions may include a code generated by a compiler or a code executable by an interpreter. The machine-readable storage medium may be provided in the form of a non-transitory storage medium. Wherein, the term "non-transitory" simply means that the storage medium is a tangible device, and does not include a signal (e.g., an electromagnetic wave), but this term does not differentiate between where data is semi-permanently stored in the storage medium and where the data is temporarily stored in the storage medium.

[0277] According to an embodiment, a method according to various embodiments of the disclosure may be included and provided in a computer program product. The computer program product may be traded as a product between a seller and a buyer. The computer program product may be distributed in the form of a machine-readable storage medium (e.g., CD-ROM), or be distributed (e.g., downloaded or uploaded) online via an application store (e.g., PlayStore™), or between two user devices (e.g., smart phones) directly. If distributed online, at least part of the computer program product may be temporarily generated or at least temporarily stored in the machine-readable storage medium, such as memory of the manufacturer's server, a server of the application store, or a relay server.

[0278] According to various embodiments, each component (e.g., a module or a program) of the above-described components may include a single entity or multiple entities, and some of the multiple entities may be separately disposed in different components. According to various embodiments, one or more of the above-described components may be omitted, or one or more other components may be added. Alternatively or additionally, a plurality of components (e.g., modules or programs) may be integrated into a single component. In such a case, according to various embodiments, the integrated component may still perform one or more functions of each of the plurality of components in the same or similar manner as they are

performed by one of the plurality of components before the integration. The integration, the operations performed by the module, the program, or another component may be carried out sequentially, in parallel, repeatedly, or heuristically, or one or more of the operations may be executed in a different order or omitted, or one or more other operations may be added.

[0279] At least one of the devices, units, components, modules, units, or the like represented by a block or an equivalent indication in the above embodiments may be physically implemented by analog and/or digital circuits including one or more of a logic gate, an integrated circuit, a microprocessor, a microcontroller, a memory circuit, a passive electronic component, an active electronic component, an optical component, and the like, and may also be implemented by or driven by software and/or firmware (configured to perform the functions or operations described herein).

[0280] Each of the embodiments provided in the above description is not excluded from being associated with one or more features of another example or another embodiment also provided herein or not provided herein but consistent with the disclosure.

[0281] The embodiments of the disclosure disclosed in the specification and the drawings provide merely specific examples to easily describe technical content according to the embodiments of the disclosure and help the understanding of the embodiments of the disclosure, not intended to limit the scope of the embodiments of the disclosure. Accordingly, the scope of various embodiments of the disclosure should be interpreted as encompassing all modifications or variations derived based on the technical spirit of various embodiments of the disclosure in addition to the embodiments disclosed herein.

Claims

1. A method of operating a first distributed unit (DU) in a wireless network system, the method comprising: monitoring radio resource allocation status information allocated to the first DU; determining changed radio resource allocation information based on the radio resource allocation status information; identifying synchronization time information for applying the changed radio resource allocation information; transmitting, to a second DU, a synchronization message comprising the synchronization time information and the changed radio resource allocation information; and performing, based on the synchronization message, transmission or reception of a message with a radio unit (RU) that is connected to the first DU and that is connected to the second DU.
2. The method of claim 1, wherein the synchronization message further comprises common signal indication information, wherein the synchronization time information comprises at least one of frame information for applying the changed radio resource allocation information or slot information for applying the changed radio resource allocation information, and wherein the common signal indication information indicates whether the first DU is to process a common signal.
3. The method of claim 1, wherein the changed radio resource allocation information comprises bitmap information indicating at least one second DU resource allocated to the second DU, and indicating at least one first DU resource allocated to the first DU.
4. The method of claim 1, wherein the changed radio resource allocation information comprises information about at least one resource region allocated to the first DU or the second DU, and wherein the information about the at least one resource region comprises information about a number of resource regions, and position information about points constituting the at least one resource region.
5. The method of claim 1, wherein the changed radio resource allocation information comprises a reference value corresponding to an integer value that separates resources allocated to the first DU and resources allocated to the second DU.
6. The method of claim 1, wherein the synchronization time information corresponds to preset time information, and wherein the preset time information comprises at least one of frame information that is preset for synchronization or slot information that is preset for synchronization.
7. The method of claim 1, wherein the synchronization time information is identified based on information about a time corresponding to a packet internet groper (PING) message being transmitted from the first DU to the second DU, and information about a time corresponding to a response message to the PING message being transmitted from the second DU to the first DU.
8. A method of operating a second distributed unit (DU) in a wireless network system, the method comprising: receiving, from a first DU, a synchronization message comprising changed radio resource allocation information and synchronization time information for applying the changed radio resource allocation information; and performing, based on the synchronization message, transmission or reception of a message with a radio unit (RU) that is connected to the first DU and that is connected to the second DU, wherein the synchronization time information for applying the changed radio resource allocation information is identified based on radio resource allocation status information allocated to the first DU.
9. The method of claim 8, wherein the synchronization message further comprises common signal indication information, wherein the synchronization time information comprises at least one of frame information for applying the changed radio resource allocation information or slot information for applying the changed radio resource allocation information, and wherein the common signal indication information indicates whether the first DU is to process a common signal.
10. The method of claim 8, wherein the changed radio resource allocation information comprises bitmap information indicating at least one second DU resource allocated to the second DU, and indicating at least one first DU resource allocated to the first DU.
11. The method of claim 8, wherein the changed radio resource allocation information comprises information about at least one resource region allocated to the first DU or the second DU, and wherein the information about the at least one resource region comprises information about a number of resource regions, and position information about points constituting the at least one resource region.
12. The method of claim 8, wherein the changed radio resource allocation information comprises a reference value corresponding to an integer value that separates resources allocated to the first DU and resources allocated to the second DU.
13. The method of claim 8, wherein the synchronization time information corresponds to preset time information, and wherein the preset time information comprises at least one of frame information that is preset for synchronization or slot information that is preset for synchronization.
14. The method of claim 8, further comprising identifying the synchronization time information for applying the changed radio resource allocation information, wherein the identifying of the synchronization time information comprises: receiving, from the first DU, a request message comprising information about a current time resource processed by the first DU; identifying the synchronization time information based on the information about the current time resource processed by the first DU and information about a current time resource processed by the second DU; and transmitting, to the first DU, the synchronization time information based on the request message.
15. A first distributed unit (DU) in a wireless network system, the first DU comprising: a transceiver; and at least one processor operatively connected to the transceiver, wherein the at least one processor is configured to monitor radio resource allocation status information allocated to the first DU, determine changed radio resource allocation information based on the radio resource allocation status information, identify synchronization time information for applying the changed radio resource allocation information, control the transceiver to transmit, to a second DU, a synchronization message comprising the synchronization time information and the changed radio resource allocation information, and perform, based on the synchronization message, transmission or reception of a message with a radio unit (RU) connected to the first DU and the second DU.
16. The first DU of claim 15, wherein the synchronization message further comprises common signal indication information, wherein the synchronization time information comprises at least one of frame information for applying the changed radio resource allocation information or slot information for applying the changed radio resource allocation information, and wherein the common signal indication information indicates whether the first DU is to process a common signal.
17. The first DU of claim 15, wherein the changed radio resource allocation information comprises bitmap information indicating at least one second DU resource allocated to the second DU, and indicating at least one first DU resource allocated to the first DU.
18. The first DU of claim 15, wherein the changed radio resource allocation information comprises information about at least one resource region

allocated to the first DU or the second DU, and wherein the information about the at least one resource region comprises information about a number resource regions, and position information about points constituting the at least one resource region.

19. The first DU of claim 15, wherein the changed radio resource allocation information comprises a reference value corresponding to an integer value that separates resources allocated to the first DU and resources allocated to the second DU.

20. The first DU of claim 15, wherein the synchronization time information corresponds to preset time information, and wherein the preset time information comprises at least one of frame information that is preset for synchronization or slot information that is preset for synchronization.
